

SESSION 2

THE MODEL THAT CHEATED

Leakage-Aware Modeling and Evidence-Based Selection

THE HIGHEST SCORE WILL BE REJECTED

MISSION AND SUCCESS CRITERIA

Four candidate models will be built under a frozen time split. One model will be deliberately allowed to use future information, then it will be exposed and excluded before a legitimate champion is selected.

4

candidate models

1

controlled leakage trap

1

eligibility-first decision

3

auditable output files

PRIMARY ARTIFACT

Leakage-Aware Model Validator and Selection Record

WHICH MODEL SHOULD BE SELECTED?

MODEL X

- Macro F1: 0.6357
- Current-time features only
- Future-year evaluation
- Eligible candidate

LEGITIMATE

MODEL Y

- Macro F1: 0.9654
- Uses future_loss_index
- Post-outcome information
- Invalid candidate

DISQUALIFIED

ELIGIBILITY IS DECIDED BEFORE PERFORMANCE IS RANKED

ACCURACY CAN HIDE COMPLETE FAILURE

Example: 1,000 risk records

900

LOW

90

MEDIUM

10

HIGH

DUMMY STRATEGY

Predict LOW
for every row

90%

accuracy

0.316

macro F1

Accuracy rewards the dominant class. Macro F1 exposes the ignored MEDIUM and HIGH classes.

PRECISION AND RECALL ANSWER DIFFERENT QUESTIONS

PRECISION

$$TP / (TP + FP)$$

Of all cases predicted as HIGH, how many were genuinely HIGH?

19 correct HIGH / 25 predicted HIGH = 0.76

RECALL

$$TP / (TP + FN)$$

Of all genuinely HIGH cases, how many were successfully detected?

19 detected HIGH / 20 actual HIGH = 0.95

Precision controls false alarms. Recall controls missed cases.

F1 FORCES PRECISION AND RECALL TO BE STRONG TOGETHER

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

The harmonic mean stays close to the weaker component.

Precision = 0.76 Recall = 0.95 F1 = 0.844

A high score is produced only when both error directions are controlled.

MACRO F1 GIVES EVERY CLASS EQUAL INFLUENCE

0.84

LOW F1

0.74

MEDIUM F1

0.84

HIGH F1

$$\text{Macro F1} = (0.84 + 0.74 + 0.84) / 3 = 0.807$$

Class frequency is not used as a weight, so minority-class failure cannot be hidden.

F1 CANNOT DETECT A SCIENTIFICALLY INVALID EXPERIMENT

F1 CAN MEASURE

precision-recall balance
for produced predictions

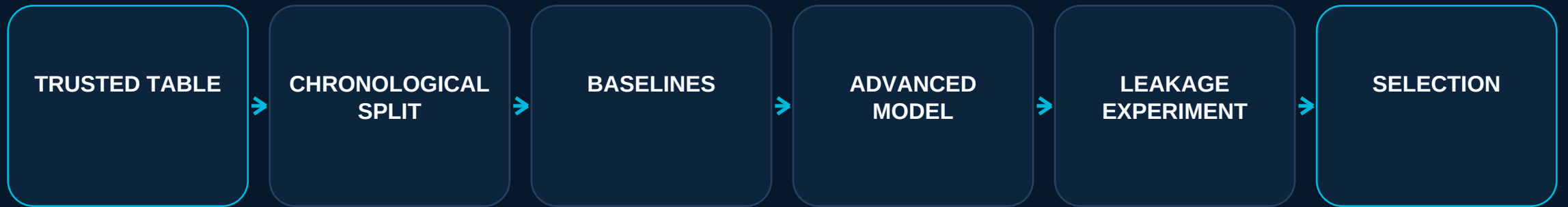
F1 CANNOT VERIFY

feature availability
or leakage-free design



A valid lower score is preferred over an invalid higher score.

SYSTEM MAP



The evaluation design answers a temporal generalization question.

FREEZE TIME BEFORE TRAINING



DEVELOP

2019-2023

Candidate models and preprocessing are fitted.



CALIBRATE

2024

Probability calibration and thresholds are reserved.



FUTURE HOLDOUT

2025

Candidates are compared under later-year conditions.

No later-year information is allowed to influence earlier preprocessing or feature design.

THE GENERALIZATION QUESTION

Can patterns learned from earlier years generalize to a later year?

The split is designed to match the question.

FOUR CANDIDATES

DUMMY

most frequent class baseline

LOGISTIC

scaled linear pipeline

XGBOOST

nonlinear corrected candidate

LEAKY XGBOOST

same family plus `future_loss_index`

BASELINES ARE NOT DECORATION

Complexity must beat a serious legitimate baseline.

The model name does not decide the winner.

LOGISTIC PIPELINE

```
numeric = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scale", StandardScaler()),
])

logistic = Pipeline([
    ("prepare", ColumnTransformer([("numeric", numeric, MODEL_FEATURES)])),
    ("model", LogisticRegression(class_weight="balanced")),
])
```

Preprocessing is fitted inside the pipeline using training data.

DELIBERATE LEAKAGE

```
leaky_features = MODEL_FEATURES + ["future_loss_index"]  
  
advanced_leaky.fit(  
    train[leaky_features],  
    y_train,  
)
```

This model is created to demonstrate a failure, not to win the comparison.

FUTURE HOLDOUT RESULTS

Candidate	Accuracy	Balanced acc.	Macro F1	Log loss
dummy	0.4083	0.3333	0.1933	21.3258
logistic	0.7417	0.7418	0.6357	0.5700
advanced_correct	0.7000	0.5030	0.4703	0.9523
advanced_leaky	0.9500	0.9706	0.9654	0.2316

The strongest numerical result is invalid because future information was used.

The logistic pipeline becomes the strongest eligible candidate.

THE LEAKAGE TRAP

0.9654 MACRO F1

INVALID

future_loss_index is a post-outcome feature and does not exist at prediction time.

A high F1 score does not certify a valid experiment.

FEATURE AVAILABILITY AUDIT

rainfall_mm

current monthly measurement - available

soil_moisture

current monthly measurement - available

disease_cases

current surveillance measurement - available

future_loss_index

post-outcome quantity - unavailable

ELIGIBILITY BEFORE RANKING

Remove invalid candidates before sorting metrics.

A prohibited candidate must not become the winner because of a larger score.

EVIDENCE-BASED SELECTION

```
legitimate = metrics[  
metrics["model"].isin([  
"dummy", "logistic", "advanced_correct"  
])  
]
```

```
selected_name = legitimate.sort_values(  
["macro_f1", "log_loss", "model"],  
ascending=[False, True, True],  
) .iloc[0]["model"]
```

Selection rule: highest legitimate macro F1, then lowest log loss.

THE SURPRISE

LOGISTIC WINS.

The corrected XGBoost candidate is more complex and scores lower. The evidence is followed.

LEGITIMATE RANKING

0.6357

LOGISTIC

SELECTED

0.4703

CORRECT XGBOOST

NOT SELECTED

0.1933

DUMMY

PERFORMANCE FLOOR

The leaky 0.9654 result is excluded before ranking begins.

WHY THIS MATTERS

A course about advanced AI should not automatically select the 'advanced' algorithm.

Advanced practice means designing the evaluation so the evidence can disagree with expectations.

METRIC CONTRACT

MACRO F1

each risk class contributes equally

BALANCED ACCURACY

recall averaged across classes

LOG LOSS

quality of predicted probabilities

CONFUSION MATRIX

class-specific error structure

HOW A CONFUSION MATRIX PRODUCES F1

HIGH-CLASS VIEW

TP = 19

FP = 6

FN = 1

TN = many

$$\text{Precision} = 19 / (19 + 6) = 0.76$$

$$\text{Recall} = 19 / (19 + 1) = 0.95$$

$$\text{F1} = 0.844$$

The confusion matrix is inspected because one Macro F1 number cannot reveal which class is failing.

MINI-PROJECT OUTPUT

model_comparison.csv

all four candidates

feature_availability_audit.csv

prediction-time availability

model_selection_record.csv

winner, rule, and exclusion reason

SIX CHECKPOINTS

1

The chronological split is created.

2

A dummy baseline and scaled logistic pipeline are built.

3

A corrected nonlinear candidate is trained.

4

A deliberate leakage model is created.

5

One shared metric table is computed.

6

The leaky candidate is excluded before selection.

Reference outcome: the logistic pipeline is selected as the strongest eligible model.

EVIDENCE TO RECORD

- 1 Why is the 0.9150 leaky score invalid?
- 2 Why is the 2025 test not used to choose thresholds later?
- 3 Why does the logistic model win on this run?
- 4 What would justify choosing XGBoost instead on another dataset?

The selection record is an auditable decision, not a preference statement.

SESSION RESULT

A legitimate product model has been selected by evidence.

The next step is to open the model and inspect what drives its decisions.